

Requirement Analysis

Solution Requirements (Functional & Non-functional)

Date	28-06-2026
Project Name	Automated Network Request Management
Team ID	
Maximum Marks	

Functional Requirements

These are the specific behaviors, features, and functions your ServiceNow system must perform based on your workflow design.

FR no.	Requirement Description	Success / Acceptance Criteria
FR-1	Automated Data Validation	The ServiceNow Service Catalog form must validate that inputs like IP addresses match standard IPv4/IPv6 regex formatting before allowing a form submission.
FR-2	Conditional Approval Routing	Flow Designer must automatically approve low-risk requests (e.g., standard ports) and route high-risk requests (e.g., SSH/RDP) to the appropriate manager or security team.
FR-3	API Integration Pipeline	Integration Hub must package approved request variables into a JSON payload and successfully push configuration changes to

		target network controllers via secure REST APIs.
FR-4	Closed-Loop Verification	The system must automatically execute a network connectivity check post-deployment to verify the change was successful before closing the ticket.
FR-5	CMDB Auto-Update & Auditing	Upon successful fulfillment, the system must write updated configuration item (CI) relationships directly to the CMDB and log an immutable audit trail on the ticket record.
FR-6	Automated Notifications	The system must send automated notifications (via email or chat webhooks) to the requestor at major lifecycle stages: submission, approval, and completion.

Non-functional Requirements

These define the system's operational attributes—how well the system performs its functions. Since you are building on the ServiceNow platform, you inherit many of these automatically, but they must still be documented.

ID	Requirement Description	Success / Acceptance Criteria

NFR-1	Performance / Speed	Automated request fulfillment (from the moment of final approval to successful API execution) must complete in less than 15 minutes for standard paths.
NFR-2	Security & Authentication	All outbound API calls from ServiceNow Integration Hub to network infrastructure must use encrypted protocols (HTTPS/SSH) and credential aliases stored securely in the platform.
NFR-3	Scalability	The workflow orchestration system must be capable of processing up to 500 concurrent automated network request transactions without causing API timeouts or platform lag.
NFR-4	Availability & Reliability	The automated request engine must achieve 99.9% uptime during operational hours, with graceful error-handling that routes failed API attempts to a manual NetOps queue.
NFR-5	Usability (UX)	Intake forms must use dynamic UI policies to show/hide fields, ensuring a minimalist layout that minimizes user input confusion and reduces incomplete submissions.